Addressing Promotion System Based on Student Data to Support Desk-to-Desk Instruction by Teaching Assistants

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ABSTRACT: Students are often believed to be solitary in programming exercise classes where they have to work on programming assignments alone. Teaching assistants (TAs) can reduce the sense of loneliness among students. Thus, in this study, we propose a system to support TAs in addressing students in desk-to-desk instruction. The proposed system uses students' data from questionnaires on student attendance, homework submission rate, review tests, and sense of loneliness to determine the priority in which to address the students. The system then automatically specifies which students should be addressed. In addition, the system presents random scenarios for TAs to address students along with hints on how the students should be addressed. The proposed system was introduced in a programming exercise class for first-year undergraduate students at a science and engineering university. Our qualitative findings showed that the system facilitated the TAs in addressing of students, which may have reduced the students' sense of loneliness.

Keywords: Teaching Assistant, Student data, Loneliness, Addressing, Scenarios, Tablet Computer

1 INTRODUCTION

Social isolation and loneliness are important aspects of the sustainable development goals established by the United Nations based on the basic principle of "No one will be left behind" (RISTEX, 2021). In general, students in programming exercise classes tend to feel isolated when they are required to do programming assignments alone. In such cases, teaching assistants (TAs) can play an important role in reducing the sense of loneliness among the students. In addition, the friendly behavior of TAs may positively impact the learning experience of students and contribute to improving their retention rate (C.O' Neal et al., 2007). For these reasons, we consider that TAs are necessary for students even today, when there exist many systems that support learning among students without TAs.

A system to support TAs is proposed based on a learning support strategy that defines which students should the TAs support during class (Imamura et al., 2020). This system lists out a number of students who require additional support from TAs based on their learning status, and the TAs themselves select the students to be guided and go to support. Our experimental results suggested that the system may encourage the TAs to address students. However, this system has two issues. First, which particular students need support is not provided. Second, the system does not provide how the TAs should support the students.

Thus, this study proposes a system to support TAs in addressing students in desk-to-desk instruction. The proposed system automatically indicates which students should be addressed. The system also presents random scenarios for addressing students and provides hints on how to address them. When TAs are equipped with these details, the number of times they address students will increase. We considered that the increase in the number of addressing and the increase in the number of

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communications between TAs and students could be an opportunity to reduce students' sense of loneliness.

2 METHODOLOGY

In this study, we propose a system that acquires data on individual students and presents scenarios for addressing each student. Figure 1 shows an overview of the system and a photo of TA holding a tablet.

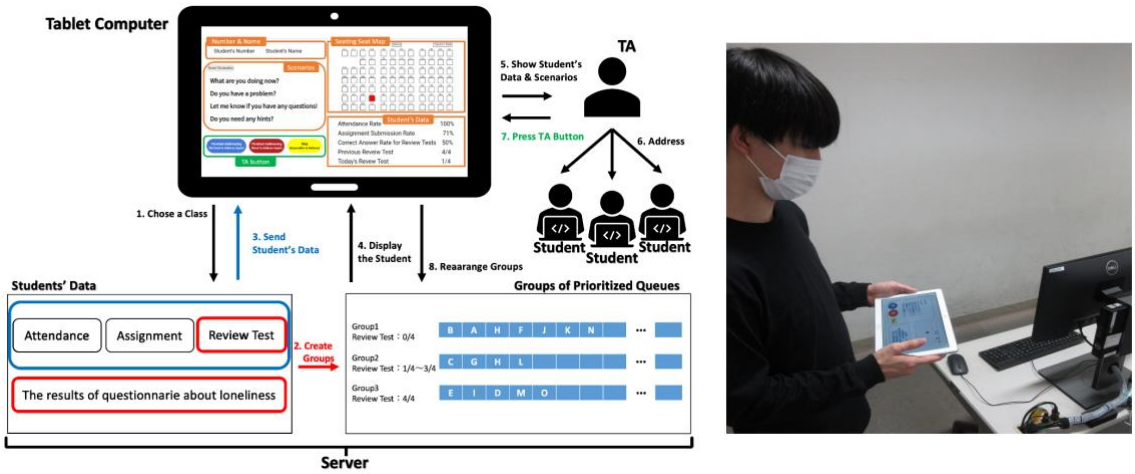

Figure 1: Overview of the method & Photo of TA holding with tablet

The upper left corner of the tablet's screen displays the identification number and the name of the student with the highest priority to be addressed. The upper right corner of the screen displays the seating chart of the students, and the seat in which the student is sitting is shown in red. The lower right portion of the screen retrieves and displays the student's attendance rate, homework submission rate, percentage of correct answers to review tests, score in the previous class review test, and score of the day's review test. Random scenarios for addressing are displayed in the left center of the screen. In the lower left part of the screen is a button to register the result of addressing that student. The blue button denotes that the TAs have approached the student and determined that the student can work on their own during the class. The red and yellow buttons denote that the TAs find it necessary to address the student again during the class. When the TA presses one of the buttons, the next student to be addressed is displayed in the same manner.

The queue is intended to determine which student should be addressed (A, B, C…, address a student). The results of a review test given to each student at the beginning of the class and the students' responses to a questionnaire about loneliness allow the TAs to make the decision to address a student. In the programming practice class in which this system is to be used, the students were tested in the review test at the beginning of the class. Based on the results of the review test, the students are categorized into three groups considering that the result of the review test indicates the student's level of understanding of the class content. For each group, the students are sorted in order of their scores on the loneliness questionnaire for feeling lonely (6-point scale of disagree to agree) as a queue (Priority Que). The first student in the group with the highest priority in the queue created above is

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displayed on the screen (student B is presented first in the figure). With respect to the reinsertion of the corresponding student into the queue, three buttons of blue, red, and yellow colors are provided and if the blue button is pressed, the student is placed at the end of the queue in the group with the lowest priority. In comparison, the student is placed at the end of the queue in the group with the original priority if the red or yellow button is pressed. Thus, we established the above rules for reinsertion into the queue, from the perspective of equality in teaching students.

The scenarios for addressing students were determined by interviewing undergraduate and graduate students with TA experience and excluding duplicate responses. We prepared 20 sets of scenarios multiplied by four of the determined scenarios and designed the system so that the scenarios would be randomly displayed on pressing the buttons to display other scenarios.

3 PRACTICE IN CLASSROOM LESSONS AND RESULTS

The system was introduced in a programming exercise class for first-year undergraduate students at a science and engineering university. The classes were taught by one teacher and two TAs, and four TAs were the subjects of analysis in two classes dealing with the same content. In class 1, the system was introduced in the 8th lesson. Our findings showed that the number of addressing times for the 8th session was 11 for TA1 and 7 for TA2. In addition, the score of questionnaires on student's sense of loneliness was improved at lesson #08 (n = 52, Ave. = 2.15, S.D. = 1.35) when compared with lesson #07 (n = 52, Ave. = 2.15, S.D. = 1.15). In class 2, the system was introduced in the 8th and 9th lessons. The number of addressing times for the 8th session was 6 for TA3 and 4 for TA4, and the score of questionnaires on student's sense of loneliness was found to be worsened at lesson #08 (n = 60, Ave. = 2.36, S.D. = 1.20) compared with lesson #07 (n = 58, Ave. = 2.36, S.D. = 1.27). For the 9th session, the number of addressing times for the 9th session was 17 for TA3 and 8 for TA4, and the score of questionnaires on student's sense of loneliness was found to be improved at lesson #09 (n = 60, Ave. = 2.10, S.D. = 0.85) compared with lesson #08 (n = 60, Ave. = 2.36, S.D. = 1.20). TA1 commented that he had previously been unable to address those students who had not completed review tests or homework because he did not know how to address such students, but this system enabled him to do so. Based on the above results, the system may have promoted TAs to talk to students, thereby reducing the students' sense of loneliness. On the other hand, the strength and timing of the effect on the sense of loneliness may differ depending on the class and TA. Therefore, it is necessary to conduct additional research needs to be conducted by increasing the number of classes and number of experiments in the future.

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